

Course Code: CS2005	Course Name: Database Systems
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Student Roll No:	Section:

Instructions:

- Return the question paper. Don't write anything on question paper, except your Roll # & Section #.
- Read each question completely before answering it. There **are 4 questions and 2 pages**.
- In case of any ambiguity, you may make assumptions. But your assumptions should not contradict any statement in the question paper.
- All the answers must be solved according to the sequence given in the question paper.
- This paper is subjective. Write the answers only on answer sheet.

Time: 60 minutes.**Max Marks:** 15**Question 1:****[2.5 marks] [CLO # 01]**

Elegance Eatery is a restaurant that has recently opened up in the neighborhood. However, they have been facing some issues due to their file – based systems. Their technical team has advised them to shift to a database based system.

1. What advantages would they have if they use a DBMS?
2. Briefly explain the three schema architecture of the DBMS with the help of a diagram.

Question 2:**[5 marks] [CLO # 01]**

Suppose that each of the following update operations is applied directly to the database state of a university's course registration system shown in Figure 01. Discuss all integrity constraints violated by each operation, if any, and the different ways of enforcing these constraints.

Student					Course		
<u>StudentID</u>	F_name	L_name	Email	Phone_no	<u>CourseID</u>	Course_name	Credits
1001	John	Smith	john@gmail.com	23215478945	C-1025	Working with JAVA	3
1002	Mary	Johnson	mary@gmail.com	45698974217	C-2145	Artificial Intelligence	3
1003	David	Williams	david@gmail.com	12457861246	C-4568	Information Security	3
1004	Emily	Davis	emily@gmail.com	12347984566			

Registration			
<u>RegistrationID</u>	StudentID	CourseID	RegistrationDate
R – 01	1001	C-1025	12 – 08 – 2023
R – 02	1003	C-4568	12 – 08 – 2023
R – 03	1002	C-1025	13 – 08 – 2023
R – 04	1001	C-2145	13 – 08 – 2023
R – 05	1004	C-4568	15 – 08 – 2023
R – 06	1002	C-4568	15 – 08 – 2023
R – 07	1003	C-2145	17 – 08 – 2023

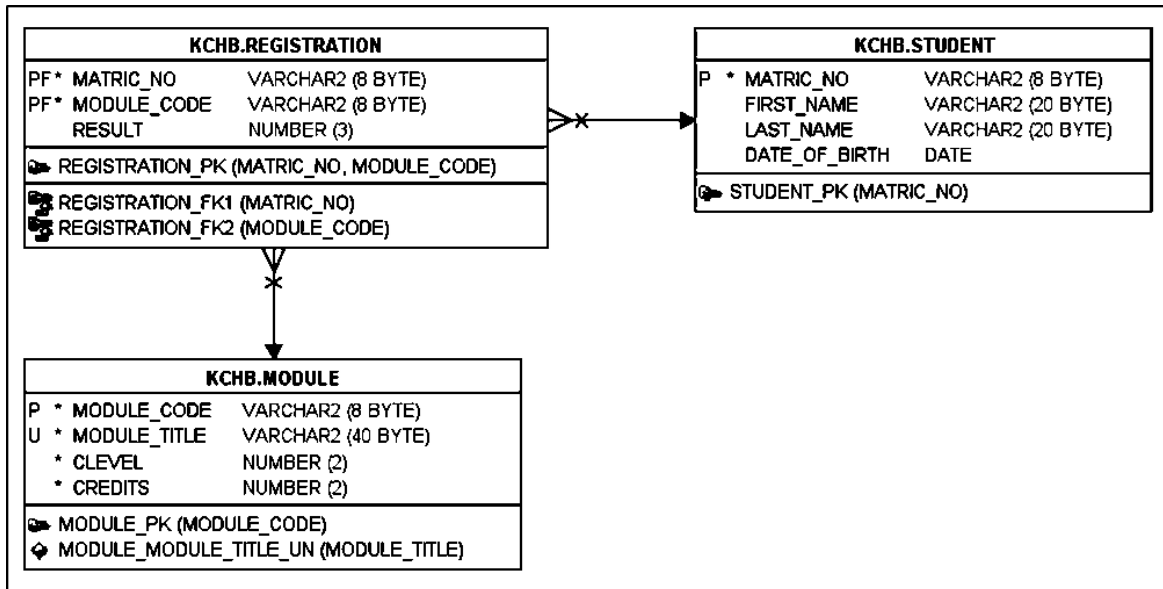
Figure 01

1. Insert <'C-2145', 'Data Mining', 3 > into Course.
2. Delete the tuple from Registration with RegistrationId = 'R – 05'.
3. Modify the StudentId attribute of the Student tuple with StudentId = 1003 to 1006.
4. Insert <'R – 08', 1003, 'C-4567', 17 – 08 – 2023 > into Registration.
5. Insert <NULL, 'James', 'Wilson', 'James@gmail.com', 25478941235 > into Student.

Question 3:**[2.5 marks] [CLO # 02]**

Write DDL for the diagram given in Figure 02.

- Apply the mentioned Key and Referential Constraints as well.
- For the Referential Constraints, specify the referential triggered actions to FK.
- The standard value of attribute Clevel (Table Module) is 20 credits; however, at Clevel 11 it is possible to have half modules worth only 10 credits.

**Figure 02****Question 4:****[5 marks] [CLO # 03]**

A Bike Company produces different bikes and bike parts. You are hired to set up a relational database for order processing. The company's order processing database has the following tables.

BRANDS (BRAND_ID , BRAND_NAME)

CATEGORIES (CATEGORY_ID, CATEGORY_NAME)

CUSTOMERS(CUSTOMER_ID,FIRST_NAME,LAST_NAME,PHONE,EMAIL,STREET,CITY,STATE,ZIP_CODE)

ORDER_ITEMS (ORDER_ID,ITEM_ID,PRODUCT_ID,QUANTITY,LIST_PRICE,DISCOUNT)

ORDERS(ORDER_ID,CUSTOMER_ID,ORDER_STATUS,ORDER_DATE,REQUIRED_DATE,SHIPPED_DATE,STORE_ID,STAFF_ID)

PRODUCTS(PRODUCT_ID,PRODUCT_NAME,BRAND_ID,CATEGORY_ID,MODEL_YEAR,LIST_PRICE)

STAFFS (STAFF_ID,FIRST_NAME,LAST_NAME,EMAIL,PHONE,ACTIVE,STORE_ID,MANAGER_ID)

STOCKS (STORE_ID,PRODUCT_ID,QUANTITY)

STORES (STORE_ID,STORE_NAME,PHONE,EMAIL,STREET,CITY,STATE,ZIP_CODE)

Using the Schema provided above, write the SQL queries for the given questions below:

1. Write a query to find the first name of customers whose name ends with 'a'.
2. Write a query to find the name of the manager of every store and the name of each staff who works in each store.
3. Write a query to find the product that is expensive compared to product with product_id 33.
4. Write a query to find the total_order(LIST_PRICE) price of each order, based on order_ids from the order_items table starting with the greatest order_id .
5. Write a query to find the name of the staff who work in the store "Baldwin Bikes" along with the name of the customer, order date, and name of the store.

Good Luck